RELATIVE DISTINGUISHED TRIANGLE ASSOCIATED TO AN OPEN COMPLEMENT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Distinguished triangle associated to an open complement (more-etale, Lemma lemma-triangle-associated-to-open, canonical tag 0GKL; collected as card L22443). cited at proof line 3905. The absolute triangle for U in X, which is pushed forward by Rf_!.
- Composition of lower shriek functors (more-etale, Lemma lemma-shriek-composition, canonical tag 0F7J). cited at proof lines 3906-3907. Identifies Rf_! Rj_! = Rg_! for g = f after j and Rf_! Ri_! = Rh_! for h = f after i.
- Rf_! for quasi-finite morphisms (more-etale, Lemma lemma-Rf-shriek-for-quasi-finite, canonical tag 0GKM). cited at proof lines 3908-3909. Gives Rj_! = j_! for the open immersion j and Ri_! = i_! = i_* for the closed immersion i.

Source reference. Original lemma 0GKL: sources/stacks-chow/tags.txt, source lines 20691–20691. Cited at proof line 3905: the absolute triangle applied to K, pushed forward by Rf_!, produces the relative triangle.

Source reference. Original lemma 0F7J: sources/stacks-chow/tags.txt, source lines 19009–19009. Cited at proof lines 3906-3907: identifies Rf_! Rj_! = Rg_! and Rf_! Ri_! = Rh_! for the restrictions g and h of f.

Source reference. Original lemma 0GKM: sources/stacks-chow/tags.txt, source lines 20692–20692. Cited at proof lines 3908-3909: rewrites the derived lower shrieks of the open and closed immersions.

Lemma 0.2. *Let $f : X \to Y$ be a finite type separated morphism of quasi-compact and quasi-separated schemes. Let U be a quasi-compact open of X with complement $Z \subset X$. Denote $g : U \to Y$ and $h : Z \to Y$ the restrictions of f. Let Λ be a ring. For K in $D^+_{tors}(X_{étale}, \Lambda)$ or $K \in D(X_{étale}, \Lambda)$ if Λ is torsion, we have a distinguished triangle*

$$Rg_!(K|_U) \to Rf_!K \to Rh_!(K|_Z) \to Rg_!(K|_U)[1]$$

in $D(Y_{étale}, \Lambda)$.

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Proof. This follows from Lemma 0GKL, the fact that $Rf_! \circ Rj_! = Rg_!$ and $Rf_! \circ Ri_!$ by Lemma 0F7J, and the fact that $Rj_! = j_!$ and $Ri_! = i_! = i_*$ by Lemma 0GKM. $\square$